

TryHackMe: Phishing Simulator

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Introduction

This project is based on my completion of the TryHackMe: Introduction to Phishing simulation. The purpose of the simulation was to provide practical experience in investigating phishing-related security incidents and understanding how Security Operations Centre (SOC) analysts respond to alerts.

Throughout the simulation, I investigated multiple security alerts, analysed evidence using Splunk, documented my findings through case notes, and managed incident tickets. The exercise allowed me to apply cybersecurity concepts in a practical environment while developing skills in threat investigation, log analysis, and incident response.

Aim of the Project

The aim of this project was to gain hands-on experience in phishing investigations and SOC operations by completing the TryHackMe simulation.

The objectives included:

- Understanding how phishing attacks are identified and investigated.
 - Learning how to analyse security events using Splunk.
 - Developing incident response and ticket management skills.
 - Recording findings and evidence throughout the investigation process.
 - Understanding the importance of SOC performance metrics such as MTBD, MTBA, and MTTR.
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What is Phishing?

Phishing is a type of cyberattack that attempts to trick users into revealing sensitive information such as usernames, passwords, financial details, or other confidential data. Attackers often use fraudulent emails that appear to come from trusted organisations in order to persuade victims to click malicious links or download harmful attachments.

Phishing remains one of the most common cyber threats because it relies on social engineering rather than technical vulnerabilities. For this reason, organisations must combine user awareness training with effective security monitoring and incident response procedures.

Tools Used

TryHackMe

TryHackMe provided the virtual environment and investigation scenarios used throughout the project. The platform guided me through the phishing simulation and presented the alerts requiring investigation.

Splunk

Splunk was used as the primary Security Information and Event Management (SIEM) tool. It allowed me to search logs, analyse security events, investigate suspicious activity, and gather evidence to support my findings.

Case Notes

Case notes were used to document observations, investigation steps, findings, and conclusions throughout the simulation. This helped maintain an organised record of the work completed.

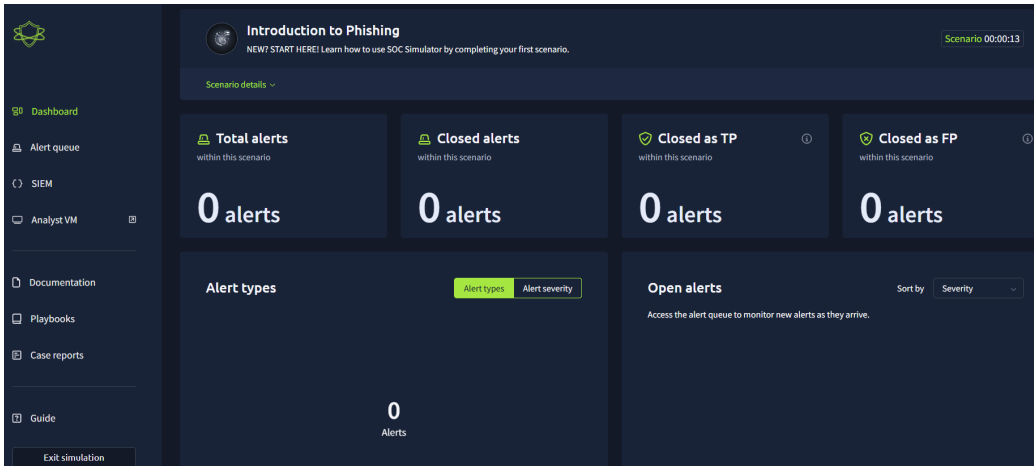
Alert Investigation and Ticket Management

As part of the simulation, I received four security alerts which required investigation and ticket management. During the exercise, I was expected to maintain key SOC performance metrics, including:

- Mean Time to Detect (MTTD)
- Mean Time to Acknowledge (MTTA)
- Mean Time to Respond (MTTR)

These metrics measure how quickly incidents are identified, reviewed, and resolved. Each alert required investigation, evidence collection, analysis, and documentation before the ticket could be closed.

Figure.1 Overview



First Ticket Investigation

Figure.2 Ticket Overview

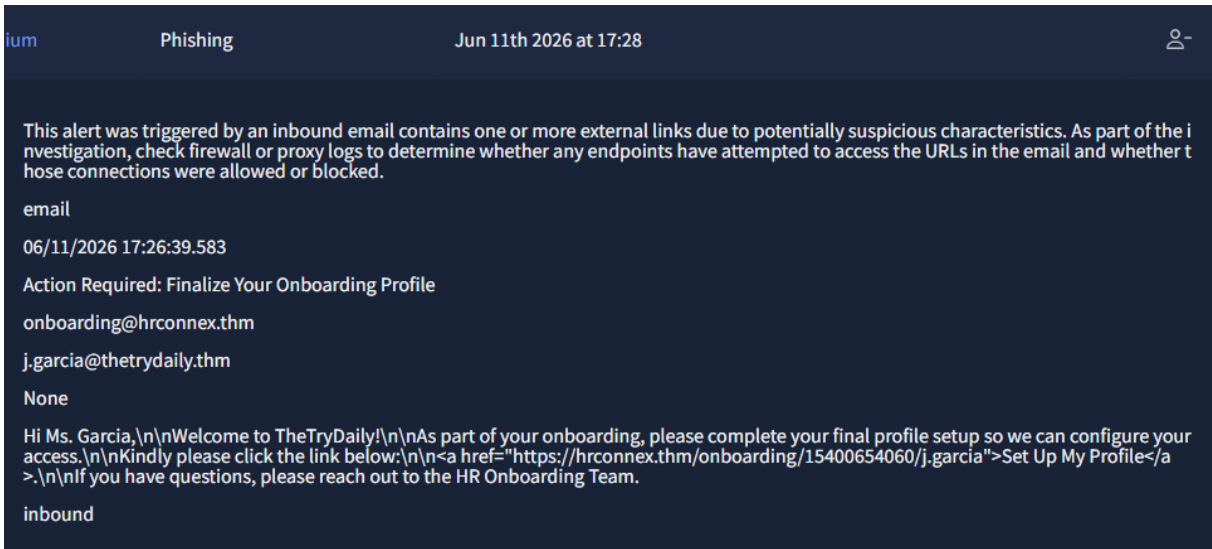


Figure 3. Splunk View



Figure 4. Case Notes

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Time of Activity:

06/11/2026 15:06:08.470

List of Related Entities:

sender: onboarding@hrconnex.thm

recipient: j.garcia@thetrydaily.thm

Link to set up: "https://hrconnex.thm/onboarding/15400654060/j.garcia"

Reason for Classifying as False Positive:

The email does not give any sign of malicious intent, upon reviewing the Splunk SIEM, everything seems to be fine. The user j.garcia is attempting to finalise her profile creation. Although the suspicious characters within the emails seem illegitimate the email contents show no harm.

The first alert was reviewed to identify any indicators of suspicious activity. I analysed the information provided, reviewed relevant logs in Splunk, and documented my findings. Based on the available evidence, I classified the alert as safe and marked it as a false positive.

Second Ticket Investigation

Figure 5. Ticket Overview

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Phishing

Jun 11th 2026 at 17:31

This alert was triggered by an inbound email contains one or more external links due to potentially suspicious characteristics. As part of the investigation, check firewall or proxy logs to determine whether any endpoints have attempted to access the URLs in the email and whether those connections were allowed or blocked.

email

06/11/2026 17:29:52.583

Your Amazon Package Couldn't Be Delivered – Action Required

urgents@amazon.biz

h.harris@thetrydaily.thm

None

Dear Customer,\n\nWe were unable to deliver your package due to an incomplete address.\n\nPlease confirm your shipping information by clicking the link below:\n\nhttp://bit.ly/3sHkX3da12340\n\nIf we don't hear from you within 48 hours, your package will be returned to sender.\n\nThank you,\n\nAmazon Delivery

inbound

Figure 6. Splunk View

```
6/11/26      { [-]
4:29:52.583 PM      attachment: None
                      content: Dear Customer,\n\nWe were unable to deliver your package due to
                      below:\n\nhttp://bit.ly/3sHkX3da12340\n\nIf we don't hear from you within 48
                      datasource: email
                      direction: inbound
                      recipient: h.harris@thetrydaily.thm
                      sender: urgents@amazon.biz
                      subject: Your Amazon Package Couldn't Be Delivered - Action Required
                      timestamp: 06/11/2026 17:29:52.583
                      }
Show as raw text
host = 1010.23.40:8989 | source = eventcollector | sourcetype = json
```

Figure 7. Case Notes

Time of activity: 06/11/2026 17:29:52.583

List of Affected Entities: sender: urgents@amazon.biz
recipient: h.harris@thetrydaily.thm
URL: \n\nhttp://bit.ly/3sHkX3da12340\n\nlf

Reason for Classifying as True Positive: There are some factors that make this a true positive - shortened URL, the urgency required from the user, as well as the fake URL with the .biz at the end

Reason for Escalating the Alert: The email itself is asking the user to enter private information in a fake website. If the user did so then, that would result in the bad actor gaining access to the user's information.

Recommended Remediation Actions: Inform users on fake URL links, they should be advised not to click URLs within emails that contain bit.ly

List of Attack Indicators: Phishing

For the second alert, this ticket contained signs of malicious intent and was an obvious phishing email. The factors such as urgency from the bad actor, a fake amazon email and a shortened URL made this obvious.

Third Ticket Investigation

Figure 8. Ticket Overview

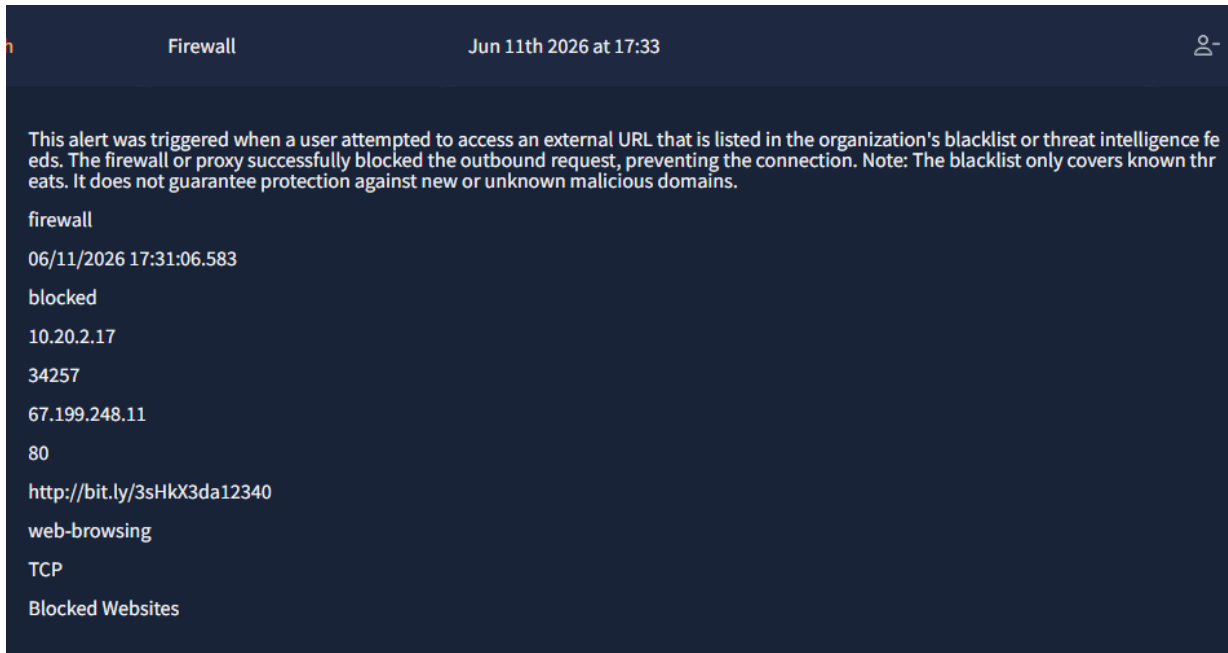


Figure 9. Splunk View

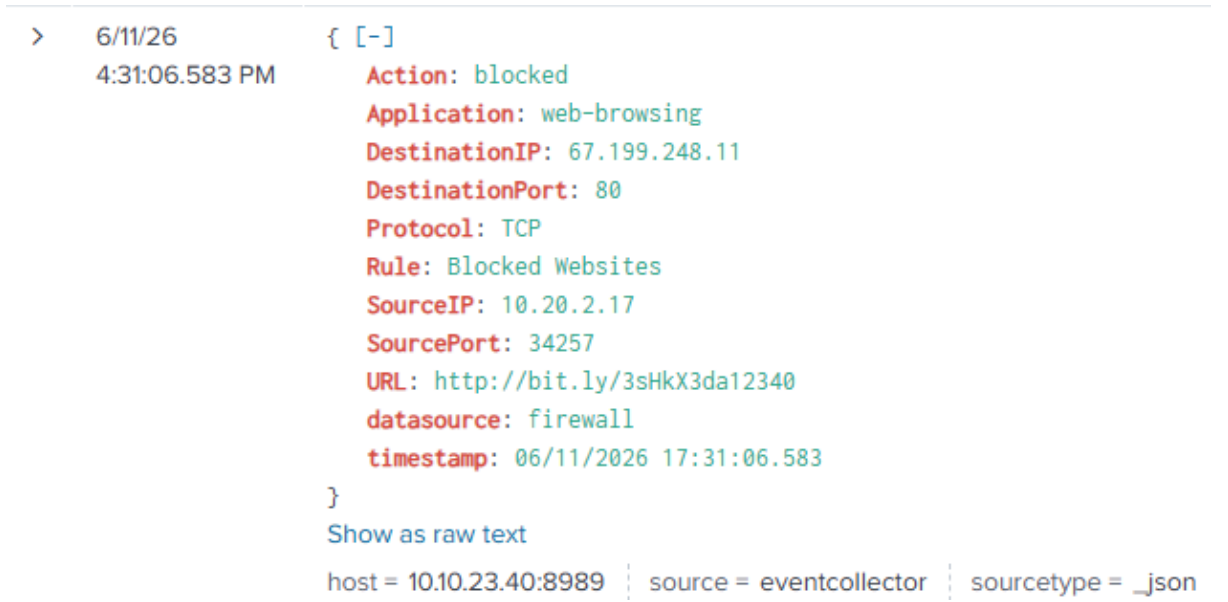


Figure 10. Case Notes

Time of activity: 06/11/2026 17:31:06.583

List of Affected Entities: The user behind the IP address - 10.20.2.17

Reason for Classifying as True Positive: The Firewall blocked a URL which was in fact within the blacklist category. The user clicked the malicious link, and the firewall did it's part by blocking the URL.

Reason for Escalating the Alert: If the firewall was not to block the URL, this would have ended the user accessing a fake link and exposing themselves to the bad actors.

Recommended Remediation Actions: Inform users on fake URL links, they should be advised not to click URLs within emails that contain bit.ly

List of Attack Indicators: Phishing

The third ticket was a form of malvertising as the user clicked on a malicious link from their google search. I was able to view this from analysing Splunk, as upon reviewing the logs I came across the user having traffic from their IP address. The site itself was blacklisted and so after clicking the link, the site got blocked.

Fourth Ticket Investigation

Figure 11. Ticket Overview

This alert was triggered by an inbound email contains one or more external links due to potentially suspicious characteristics. As part of the investigation, check firewall or proxy logs to determine whether any endpoints have attempted to access the URLs in the email and whether those connections were allowed or blocked.

email

06/11/2026 17:32:10.583

Unusual Sign-In Activity on Your Microsoft Account

no-reply@microsoftsupport.co

c.allen@thetrydaily.thm

None

Hi C.Allen,\n\nWe detected an unusual sign-in attempt on your Microsoft account.\n\nLocation: Lagos, Nigeria\n\nIP Address: 102.89.222.143\n\nDate: 2025-01-24 06:42\n\nIf this was not you, please secure your account immediately to avoid unauthorized access.\n\n<https://microsoftsupport.co/login>>Review Activity\n\nThank you,\n\nMicrosoft Account Security Team

inbound

Figure 12. Splunk View

Time	Event
6/11/26 4:32:10.583 PM	<pre>{ [-] attachment: None content: Hi C.Allen,\n\nWe detected an unusual sign-in attempt secure your account immediately to avoid unauthorized access.\n\ datasource: email direction: inbound recipient: c.allen@thetrydaily.thm sender: no-reply@microsoftsupport.co subject: Unusual Sign-In Activity on Your Microsoft Account timestamp: 06/11/2026 17:32:10.583 }</pre>

Figure 13. Case Notes

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Time of activity: 06/11/2026 17:32:10.583

List of Affected Entities: sender: no-reply@microsoftsupport.co
recipient: c.allen@thetrydaily.thm

Reason for Classifying as True Positive: The email within the sender contains suspicious characters. They are tricking the user into thinking they are from the Microsoft Support Team.

Reason for Escalating the Alert: Within the email, are characters that are malicious for instance "https://m1crosoftsupport.co/login".

Recommended Remediation Actions: Give users training on how to spot fake emails that come across as real. As well as this, make sure they do not click on links that contain suspicious characters.

List of Attack Indicators: Phishing

Alert four had red flags written all over it, as the user seems to have their account compromised. In this matter, making sure the user's account was locked was vital as well as enforcing the user to create new credentials. On top of this, the sender's email address and IP address needs to be blocked, so further attacks from them do not occur.

Figure 14. Simulation Completed

prakash05

Victory! Security breach prevented!

You passed the scenario by identifying all true positive alerts. Your MTTR and dwell time were quicker than average, but your true positive rate was 60%, worse than previous runs. "Phishing" alerts took longer to close.

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What I Learned

Completing this simulation helped me develop both technical and analytical cybersecurity skills.

I gained practical experience in identifying phishing indicators, investigating security alerts, using Splunk to analyse log data, and documenting findings through case notes and ticket updates. The exercise also improved my understanding of incident response procedures and the importance of maintaining key SOC metrics such as MTTD, MTTA, and MTTR.

Perhaps most importantly, the simulation demonstrated how phishing attacks are investigated in real-world environments and highlighted the role of SOC analysts in protecting organisations from cyber threats.

References

TryHackMe – Introduction to Phishing

Splunk Documentation

National Cyber Security Centre (NCSC) – Phishing Guidance

OWASP Security Awareness Resources